Strava Fitness Tracker Data Analysis Report

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1. Introduction

Smart fitness trackers collect large amounts of health-related data such as activity levels, sleep patterns, calorie expenditure, and body metrics. Analyzing this data helps businesses understand user behavior and improve product engagement and marketing strategies.

The objective of this project is to analyze smart fitness tracker data to identify patterns in **physical activity, sleep behavior, calorie burn, and BMI trends**.

This analysis was conducted using **Python for data cleaning and exploratory analysis, SQL for querying and structured analysis, and Power BI for interactive dashboards**.

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2. Business Objective

The business objective of this project is to:

- Analyze smart fitness tracker data to understand **user behavior patterns**
- Identify patterns in **activity levels, sleep duration, calorie burn, and BMI**
- Understand **user engagement with fitness trackers**
- Detect **health risks based on BMI and sedentary behavior**
- Provide **data-driven insights to improve marketing strategies and product features**

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3. Dataset Selection

From the complete dataset collection, the following **5 datasets** were selected because they directly support the business objective.

Dataset	Description
dailyActivity_merged	Contains daily steps, activity minutes, calories burned
sleepDay_merged	Contains sleep duration and sleep sessions
weightLogInfo_merged	Contains weight and BMI information
dailySteps_merged	Contains step counts daily
dailyCalories_merged	Contains calorie burnt daily

These datasets provide insights into **activity, sleep, calorie expenditure, and BMI**, which are essential for understanding user fitness behavior.

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4. Data Cleaning and Processing (Python)

Python was used to extract, clean, and prepare the datasets before analysis.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn

Data Cleaning Steps

1. Imported datasets using Pandas.
2. Checked dataset structure using .info() and .describe().
3. Removed duplicate records.
4. Handled missing values.
5. Converted date and time columns to proper datetime format.
6. Selected relevant columns for analysis.

Example Cleaning Code

```
import pandas as pd

activity = pd.read_csv("dailyActivity_merged.csv")
sleep = pd.read_csv("sleepDay_merged.csv")
weight = pd.read_csv("weightLogInfo_merged.csv")

activity.drop_duplicates(inplace=True)
sleep.drop_duplicates(inplace=True)

sleep['SleepDay'] = pd.to_datetime(sleep['SleepDay'])
```

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5. Exploratory Data Analysis (EDA)

EDA was performed to understand the distribution, patterns, and relationships within the datasets.

The analysis focused on:

- Activity intensity
- Daily steps
- Sleep duration
- Calories burned

- BMI distribution

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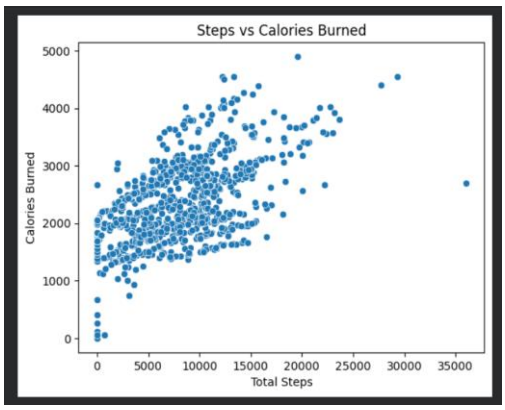
6. Python Visualizations and Insights

Chart 1: Steps vs Calories Burned



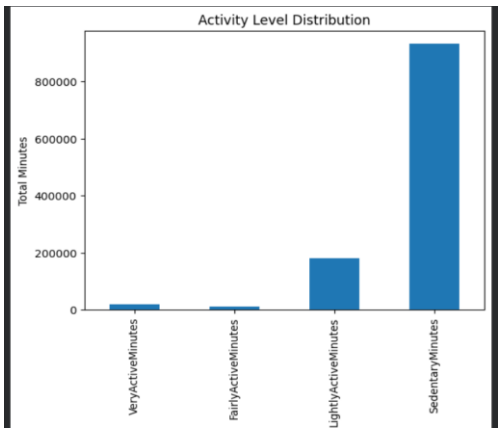
Insight

- Users who walk more steps tend to burn more calories.
- Shows a strong positive relationship between activity and calorie burn.

Impact

Positive: Encourages activity tracking
Negative: Some users show low activity but high calorie intake.

Chart 2: Activity Level Distribution



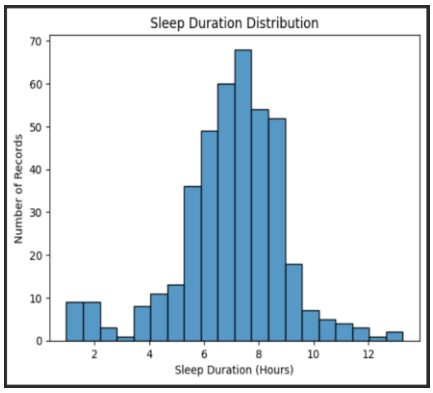
Insight

- Majority of time is spent in **sedentary activities**.
- Very active minutes are significantly lower.

Impact

Negative: Indicates low physical activity among users

Chart 3: Sleep Duration Distribution



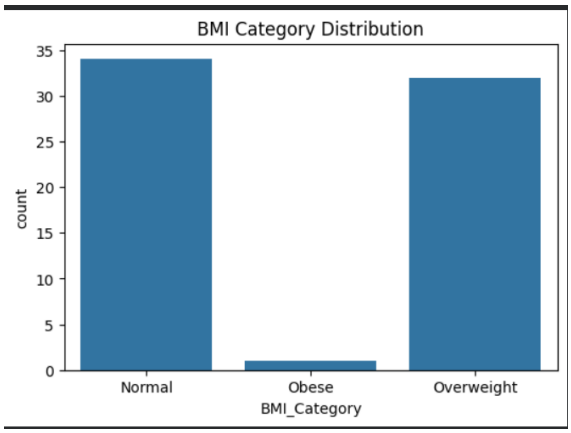
Insight

- Most users sleep between **6–8 hours**, which is considered healthy.
- Some users sleep less than 5 hours.

Impact

Positive: Many users maintain healthy sleep patterns
Negative: Short sleep durations for some users.

Chart 4: BMI Distribution



Insight

- Majority of users fall in the **normal BMI range**.
- Some users fall in the **overweight category**, indicating potential health risks.
- BMI distribution is **slightly right-skewed**, meaning some users have higher BMI values.

Impact

Positive: Many users Fall in normal BMI Category
Negative: Some percent of the users fall under Obese and overweight category so need to focus on activity

7. Exporting Cleaned Dataset

After cleaning and analysing, the dataset was exported.

This dataset was then used for **SQL analysis and Power BI dashboards**.

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8. SQL Data Processing

The cleaned dataset was imported into **MySQL** for further analysis.

SQL Cleaning Steps

- Created tables
- Changed data types
- Checked for null values
- Ensured ID columns were numeric

Example:

```
ALTER TABLE fitness_data
MODIFY Id BIGINT;
```

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9. SQL Queries and Tabular Results

Query 1: Average Daily Steps

```
SELECT Id, AVG(TotalSteps) AS AvgSteps
FROM fitness_data
GROUP BY Id;
```

Query 2: check duplicates

```
SELECT Id, ActivityDate, COUNT(*)
FROM daily_activity
GROUP BY Id, ActivityDate
HAVING COUNT(*) > 1;
```

Query 3: Average Calories Burned

```
SELECT Id, AVG(Calories) AS AvgCalories
FROM fitness_data
GROUP BY Id;
```

Query 4: BMI Category Count

```
SELECT
CASE
WHEN BMI < 18.5 THEN 'Underweight'
WHEN BMI BETWEEN 18.5 AND 24.9 THEN 'Normal'
WHEN BMI BETWEEN 25 AND 29.9 THEN 'Overweight'
```

```
ELSE 'Obese'
END AS BMI_Category,
COUNT(*) AS Users
FROM weightloginfo
GROUP BY BMI_Category;
```

Query 5: Most Active Users

```
SELECT Id, SUM(TotalSteps) AS TotalSteps
FROM fitness_data
GROUP BY Id
ORDER BY TotalSteps DESC;
```

These results were presented in **tabular format**.

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10. Power BI Dashboard Development

Power BI was connected to MySQL to create **interactive dashboards**.

Connection Method:

Power BI → Get Data → MySQL Database → Select Table

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11. Dashboard 1: User Engagement

Visuals

- Total Users
- Total Active days
- Activity Level Distribution
- Daily Activity Pattern
- Steps vs Calories

Insights

Positive Impact

- Many users actively track their daily steps.

Negative Impact

- High sedentary time indicates low engagement with physical activities.

Business Solution

- Introduce **daily step challenges** to increase activity.

12. Dashboard 2: Sleeping Pattern

Visuals

- Average Sleep Duration
- Sleep Duration Distribution
- Sleep vs Activity
- Sleep Efficiency based on actual asleep time
- Steps proportional to good sleep

Insights

Positive Impact

- Majority of users maintain healthy sleep hours.

Negative Impact

- Some users show poor sleep patterns.

Business Solution

- Introduce **sleep tracking reminders and insights notifications**.

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13. Dashboard 3: Weight and BMI Index

Visuals

- BMI Distribution
- Weight Distribution
- BMI Categories
- Weight vs Activity
- BMI vs Calories

Insights

Positive Impact

- Majority of users fall in **normal BMI range**.

Negative Impact

- Some users are **overweight**, indicating potential health risks.

BMI distribution is **slightly right-skewed**, showing higher BMI values for some users.

Business Solution

- Provide **personalized workout and diet recommendations**.

14. Key Business Insights

1. Most users spend significant time in sedentary activity.
2. Higher step counts correlate with higher calorie burn.
3. Most users maintain normal BMI levels.
4. Some users show potential health risks due to higher BMI.
5. Sleep duration varies significantly among users.

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15. Recommendations

Based on the analysis, the following strategies are recommended:

- Introduce **fitness challenges to increase user engagement**.
- Send **sleep reminders and sleep quality insights**.
- Provide **personalized fitness recommendations**.
- Implement **BMI-based health guidance**.
- Encourage users to reduce sedentary time through notifications.